



GSFA Online

Updated up to the 48th Session of the Codex Alimentarius Commission (2025)

FOOD CATEGORY DETAILS

Smoked, dried, fermented, and/or salted fish and fish products, including mollusks, crustaceans, and echinoderms (09.2.5)

Description:

Smoked fish are usually prepared from fresh deep frozen or frozen fish that are dried directly or after boiling, with or without salting, by exposing the fish to freshly-generated sawdust smoke. Dried fish are prepared by exposing the fish to sunlight or drying directly or after boiling in a special installation; the fish may be salted prior to drying. Salted fish are either rubbed with salt or placed in a salt solution. This manufacturing process is different from that described in food category 09.3 for marinated and pickled fish. Cured fish is prepared by salting and then smoking fish.¹ Examples include: salted anchovies, shrimp, and shad; smoked chub, cuttlefish and octopus; fish ham; dried and salted species of the Gadidae species; smoked or salted fish paste and fish roe; cured and smoked sablefish, shad, and salmon; dried shellfish, dried bonito (katsuobushi), and boiled, dried fish (niboshi).

This page provides information on the food additive provisions that are acceptable for use in foods conforming to the food category.

This food category is listed in the **Annex to Table 3**. Unless specifically indicated below, food additive provisions implied by **Table 3** do not automatically apply to this category.

GSFA Provisions for Food Category 09.2.5

INS No.	Food Additive or Group	Max Level	Notes	Defined In
950	Acesulfame potassium	200 mg/kg	 Note XS95  Note XS189  Note 144  Note XS222  Note XS191  Note XS166  Note XS312  Note XS244  Note XS190  Note XS36  Note XS167  Note 478  Note XS315  Note XS292  Note XS236  Note 188  Note XS92  Note XS311  Note XS165	09.2
260	Acetic acid, glacial	GMP	 Note XS244  Note XS167  Note XS189  Note 437  Note XS222  Note XS236	09.2.5
472a	Acetic and fatty acid esters of glycerol	GMP	 Note XS244  Note 300  Note XS167  Note XS189  Note XS222  Note XS236  Note XS311	09.2.5
1414	Acetylated distarch phosphate	GMP	 Note XS244  Note 300	09.2.5

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	
969	Advantame	3 mg/kg	Note 144	09.2
406	Agar	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
400	Alginic acid	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note 332 Note XS311	09.2.5
129	Allura red AC	300 mg/kg	Note XS244 Note 382 Note XS167 Note XS189 Note XS222 Note XS236	09.2.5
160b(i)	Annatto extracts, bixin-based	10 mg/kg	Note XS244 Note 382 Note XS167 Note XS189 Note XS222 Note 8 Note XS236	09.2.5
300	Ascorbic acid, L-	GMP	Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
951	Aspartame	300 mg/kg	Note XS95 Note XS189 Note 144 Note XS222 Note XS191 Note XS166 Note XS312 Note XS244 Note XS190 Note XS36 Note XS167 Note 478 Note XS315 Note XS292 Note 191 Note XS236 Note XS92 Note XS311 Note XS165	09.2
962	Aspartame-acesulfame salt	200 mg/kg	Note XS95 Note 144 Note XS189 Note XS222 Note XS191 Note XS166 Note XS312 Note XS244	09.2

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			Note XS190 Note XS36 Note XS167 Note XS315 Note XS292 Note XS236 Note XS92 Note 113 Note XS311 Note XS165	
	BENZOATES	200 mg/kg	Note XS167 Note XS189 Note XS222 Note XS236 Note 333 Note 13 Note 121	09.2.5
320	Butylated hydroxyanisole (BHA)	200 mg/kg	Note XS244 Note XS167 Note 15 Note XS189 Note XS222 Note XS236 Note XS311 Note 196	09.2.5
321	Butylated hydroxytoluene (BHT)	200 mg/kg	Note XS244 Note XS167 Note 15 Note XS189 Note XS222 Note XS236 Note XS311 Note 196	09.2.5
	CAROTENES, BETA-	100 mg/kg	Note XS95 Note XS189 Note XS222 Note XS191 Note XS312 Note XS244 Note XS190 Note XS36 Note XS167 Note 341 Note XS315 Note XS292 Note 344 Note XS236 Note XS92 Note XS311 Note XS165	09.2
	CHLOROPHYLLS AND CHLOROPHYLLINS, COPPER COMPLEXES	200 mg/kg	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
170(i)	Calcium carbonate	GMP	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
509	Calcium chloride	GMP	Note XS244 Note 300 Note XS167 Note XS189	09.2.5

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			Note XS222 Note XS236 Note XS311	
327	Calcium lactate	GMP	Note XS244 Note XS167 Note XS189 Note 437 Note XS222 Note XS236	09.2.5
161g	Canthaxanthin	15 mg/kg	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note 22 Note XS311	09.2.5
150c	Caramel III - ammonia caramel	30,000 mg/kg	Note XS95 Note XS189 Note XS222 Note XS191 Note XS166 Note XS312 Note XS244 Note XS190 Note XS36 Note XS167 Note XS315 Note XS292 Note XS236 Note XS92 Note XS311 Note XS165	09.2
150d	Caramel IV - sulfite ammonia caramel	30,000 mg/kg	Note 95 Note XS95 Note XS189 Note XS222 Note XS191 Note XS166 Note XS312 Note XS244 Note XS190 Note XS36 Note XS167 Note XS315 Note XS292 Note XS236 Note XS92 Note XS311 Note XS165	09.2
290	Carbon dioxide	GMP	Note XS244 Note 382 Note XS167 Note XS189 Note XS222 Note 59 Note XS236	09.2.5
120	Carmines	300 mg/kg	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note 22 Note XS311 Note 178	09.2.5
160e	Carotenal, beta-apo-8'-	100 mg/kg	Note 439 Note XS95	09.2

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			Note XS189 Note XS222 Note XS191 Note XS312 Note XS244 Note XS190 Note XS36 Note XS167 Note XS315 Note XS292 Note XS236 Note 336 Note XS92 Note XS311 Note XS165	
160a(ii)	beta-Carotenes, vegetable	100 mg/kg	Note 439 Note XS95 Note XS189 Note XS222 Note XS191 Note XS312 Note XS244 Note XS190 Note XS36 Note XS167 Note 341 Note XS315 Note XS292 Note 344 Note XS236 Note 336 Note XS92 Note XS311 Note XS165	09.2
407	Carrageenan	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note 332 Note XS311	09.2.5
330	Citric acid	GMP	Note XS167 Note XS189 Note 437 Note XS222 Note XS236	09.2.5
472c	Citric and fatty acid esters of glycerol	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
627	Disodium 5'-guanylate	GMP	Note XS244 Note XS167 Note 29 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
631	Disodium 5'-inosinate	GMP	Note XS244 Note XS167 Note 29 Note XS189 Note XS222 Note XS236	09.2.5

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
635	Disodium 5'-ribonucleotides	GMP	Note XS311 Note XS244 Note XS167 Note 29 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
143	Fast green FCF	100 mg/kg	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
297	Fumaric acid	GMP	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
422	Glycerol	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
163(ii)	Grape skin extract	1,000 mg/kg	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note 22 Note XS311	09.2.5
412	Guar gum	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
414	Gum arabic (Acacia gum)	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note 332 Note XS311	09.2.5
463	Hydroxypropyl cellulose	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note 332 Note XS311	09.2.5
464	Hydroxypropyl methyl cellulose	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note 332 Note XS311	09.2.5

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
1440	Hydroxypropyl starch	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
	IRON OXIDES	250 mg/kg	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note 22 Note XS311	09.2.5
425	Konjac flour	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note 332 Note XS311	09.2.5
270	Lactic acid, L-, D- and DL-	GMP	Note XS244 Note 382 Note XS167 Note XS189 Note XS222 Note XS236	09.2.5
472b	Lactic and fatty acid esters of glycerol	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
243	Lauric arginate ethyl ester	200 mg/kg	Note XS189 Note XS222 Note XS236 Note 333	09.2.5
322(i)	Lecithin	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
504(i)	Magnesium carbonate	GMP	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
511	Magnesium chloride	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
528	Magnesium hydroxide	GMP	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
504(ii)	Magnesium hydroxide carbonate		GMP - Note XS244 - Note XS167 - Note XS189 - Note XS222 - Note XS236 - Note XS311	09.2.5
296	Malic acid, DL-		GMP - Note XS244 - Note XS167 - Note XS189 - Note XS222 - Note XS236 - Note XS311	09.2.5
965(i)	Maltitol		GMP Note 489	09.2.5
965(ii)	Maltitol syrup		GMP Note 489	09.2.5
421	Mannitol		GMP - Note XS244 - Note 300 - Note XS167 - Note XS189 - Note XS222 - Note XS236 - Note XS311	09.2.5
461	Methyl cellulose		GMP - Note XS244 - Note 300 - Note XS167 - Note XS189 - Note XS222 - Note XS236 - Note 332 - Note XS311	09.2.5
465	Methyl ethyl cellulose		GMP - Note XS244 - Note 300 - Note XS167 - Note XS189 - Note XS222 - Note XS236 - Note XS311	09.2.5
460(i)	Microcrystalline cellulose (Cellulose gel)		GMP - Note XS244 - Note 300 - Note XS167 - Note XS189 - Note XS222 - Note XS236 - Note 332 - Note XS311	09.2.5
471	Mono- and di-glycerides of fatty acids		GMP - Note XS244 - Note 300 - Note XS167 - Note XS189 - Note XS222 - Note XS236 - Note XS311	09.2.5
621	Monosodium L-glutamate		GMP - Note XS244 - Note XS167 - Note 29 - Note XS189 - Note XS236 - Note XS311 - Note 313	09.2.5
941	Nitrogen		GMP - Note XS244 - Note 382 - Note XS167 - Note XS189 - Note XS222 - Note 59 - Note XS236	09.2.5

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
1404	Oxidized starch	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
	PHOSPHATES	2,200 mg/kg	Note XS244 Note XS167 Note 420 Note 33 Note XS189 Note 334 Note XS236 Note 413 Note XS311	09.2.5
440	Pectins	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
124	Ponceau 4R (Cochineal red A)	100 mg/kg	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note 22 Note XS311	09.2.5
501(i)	Potassium carbonate	GMP	Note XS244 Note XS167 Note 230 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
508	Potassium chloride	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
332(i)	Potassium dihydrogen citrate	GMP	Note XS95 Note XS189 Note XS222 Note XS191 Note XS312 Note 253 Note XS244 Note XS190 Note XS36 Note XS167 Note 391 Note XS315 Note XS292 Note XS236 Note XS92 Note XS311	09.2
326	Potassium lactate	GMP	Note XS244 Note 382 Note XS167 Note XS189 Note XS222 Note XS236	09.2.5

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
460(ii)	Powdered cellulose	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note 332 Note XS311	09.2.5
407a	Processed eucheuma seaweed (PES)	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note 332 Note XS311	09.2.5
310	Propyl gallate	100 mg/kg	Note XS244 Note XS167 Note 15 Note XS189 Note XS222 Note XS236 Note XS311 Note 196	09.2.5
1204	Pullulan	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
	RIBOFLAVINS	GMP	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note 22 Note XS311	09.2.5
	SORBATES	1,000 mg/kg	Note XS189 Note XS222 Note 20 Note XS236 Note 42	09.2.5
	STEVIOL GLYCOSIDES	165 mg/kg	Note 489 Note 208 Note 26	09.2.5
	SULFITES	30 mg/kg	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311 Note 44	09.2.5
470(i)	Salts of myristic, palmitic and stearic acids with ammonia, calcium, potassium and sodium	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
470(ii)	Salts of oleic acid with calcium, potassium and sodium	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236	09.2.5

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			 Note XS311	
350(ii)	Sodium DL-malate	GMP	 Note XS244  Note XS167  Note XS189  Note XS222  Note XS236  Note XS311	09.2.5
262(i)	Sodium acetate	GMP	 Note XS244  Note XS167  Note XS189  Note XS222  Note XS236  Note XS311	09.2.5
401	Sodium alginate	GMP	 Note XS244  Note 300  Note XS167  Note XS189  Note XS222  Note XS236  Note 332  Note XS311	09.2.5
301	Sodium ascorbate	GMP	 Note XS244  Note XS167  Note XS315  Note XS189  Note 437  Note XS222  Note XS236  Note XS191  Note XS312  Note XS92  Note 392  Note 307	09.2
500(i)	Sodium carbonate	GMP	 Note XS244  Note XS167  Note XS189  Note XS222  Note XS236  Note XS311	09.2.5
466	Sodium carboxymethyl cellulose (Cellulose gum)	GMP	 Note XS244  Note 300  Note XS167  Note XS189  Note XS222  Note XS236  Note 332  Note XS311	09.2.5
331(i)	Sodium dihydrogen citrate	GMP	 Note XS95  Note XS189  Note XS222  Note XS191  Note XS312  Note 253  Note XS244  Note XS190  Note XS36  Note XS167  Note 391  Note XS315  Note XS292  Note XS236  Note XS92  Note XS311	09.2
316	Sodium erythorbate (Sodium isoascorbate)	GMP	 Note XS244 Note 382 Note XS167	09.2.5

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			Note XS189 Note XS222 Note XS236	
365	Sodium fumarates	GMP	Note XS244 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
576	Sodium gluconate	GMP	Note XS95 Note XS189 Note XS222 Note XS191 Note XS166 Note XS312 Note XS244 Note XS190 Note XS36 Note XS167 Note XS315 Note XS292 Note XS236 Note XS92 Note XS311 Note XS165	09.2
325	Sodium lactate	GMP	Note XS244 Note XS167 Note XS189 Note 437 Note XS222 Note XS236	09.2.5
420(i)	Sorbitol	GMP		09.2.5
420(ii)	Sorbitol syrup	GMP		09.2.5
110	Sunset yellow FCF	100 mg/kg	Note XS244 Note 382 Note XS167 Note XS189 Note XS222 Note XS236	09.2.5
	TARTRATES	200 mg/kg	Note XS244 Note 382 Note XS167 Note XS189 Note XS222 Note 128 Note XS236 Note 45	09.2.5
417	Tara gum	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222 Note XS236 Note XS311	09.2.5
102	Tartrazine	100 mg/kg	Note XS244 Note 382 Note XS167 Note XS189 Note XS222 Note XS236	09.2.5
413	Tragacanth gum	GMP	Note XS244 Note 300 Note XS167 Note XS189 Note XS222	09.2.5

INS No. ▼	Food Additive or Group	Max Level ▼	Notes ▼	Defined In ▼
			☐ Note XS236 ☐ Note XS311	
333(iii)	Tricalcium citrate	GMP	☐ Note XS95 ☐ Note XS189 ☐ Note XS222 ☐ Note XS191 ☐ Note XS166 ☐ Note XS312 ☐ Note XS244 ☐ Note XS190 ☐ Note XS36 ☐ Note XS167 ☐ Note XS315 ☐ Note XS292 ☐ Note XS236 ☐ Note XS92 ☐ Note XS311 ☐ Note XS165	09.2
332(ii)	Tripotassium citrate	GMP	☐ Note XS95 ☐ Note XS189 ☐ Note XS222 ☐ Note XS191 ☐ Note XS312 ☐ Note 253 ☐ Note XS244 ☐ Note XS190 ☐ Note XS36 ☐ Note XS167 ☐ Note 391 ☐ Note XS315 ☐ Note XS292 ☐ Note XS236 ☐ Note XS92 ☐ Note XS311	09.2
331(iii)	Trisodium citrate	GMP	☐ Note XS95 ☐ Note XS189 ☐ Note XS222 ☐ Note XS191 ☐ Note XS312 ☐ Note 253 ☐ Note XS244 ☐ Note XS190 ☐ Note XS36 ☐ Note XS167 ☐ Note 391 ☐ Note XS315 ☐ Note XS292 ☐ Note XS236 ☐ Note XS92 ☐ Note XS311	09.2
415	Xanthan gum	GMP	☐ Note XS244 ☐ Note 300 ☐ Note XS167 ☐ Note XS189 ☐ Note XS222 ☐ Note XS236 ☐ Note XS311	09.2.5

Items in uppercase (e.g. **PHOSPHATES**) refer to food additive groups.

1. Food Chemistry, H.-D. Belitz & W. Grosch, Springer-Verlag, Heidelberg, 1987, pp. 464-68.

